

HYPERPARAMETER TUNING OF DATA MINING MODELS

Systematic Optimization of Algorithmic Hyperparameters for Predictive Accuracy & High-Dimensional Pattern Discovery

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1. FUNDAMENTALS & MATHEMATICAL FORMULATION

THEORY

In machine learning and data mining, models possess two distinct parameter tiers: **Model Parameters ($\mathbf{w}$)**, learned automatically during training via gradient descent or closed-form solutions, and **Hyperparameters (θ)**, structural configurations set *prior* to training that govern algorithm learning dynamics, capacity, and model topology.

Bilevel Optimization Objective

$$\theta^* = \arg \min_{\theta \in \Theta} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim D_{\text{val}}} [L(f(\mathbf{x}; \mathbf{w}^*(\theta)), \mathbf{y})]$$
$$\text{subject to: } \mathbf{w}^*(\theta) = \arg \min_{\mathbf{w}} L_{\text{train}}(f(\mathbf{x}; \mathbf{w}), \mathbf{y}_{\text{train}})$$

Hyperparameter tuning systematically searches the multi-dimensional configuration space Θ to discover the parameter tuple θ^* that minimizes generalization error on unseen validation data without incurring overfitting.

2. SYSTEMATIC SEARCH STRATEGIES

ALGORITHMS

- Grid Search (Exhaustive):** Evaluates the Cartesian product of predefined discrete values. Scales as $O(M^d)$ (curse of dimensionality); wastes compute on insensitive axes.
- Random Search (Bergstra & Bengio):** Samples trials uniformly at random. Dramatically superior in practice because most ML models exhibit low *effective dimensionality* (only 1–2 parameters dominate performance).
- Bayesian Optimization:** Fits a probabilistic surrogate model (Gaussian Process / TPE) to historical trials, using an Acquisition Function (Expected Improvement $El(x)$) to balance exploration vs. exploitation.
- Successive Halving & Hyperband:** Allocates dynamic early-stopping budgets; evaluates many configurations on small data subsets and doubles resources only for top performers.

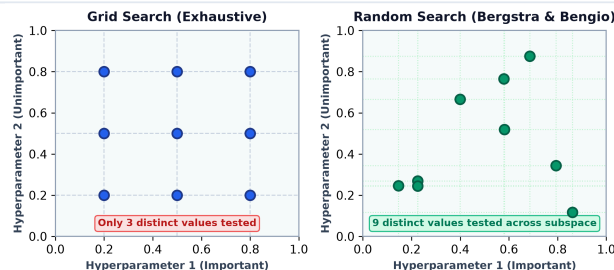


Figure 1: Grid Search vs Random Search across 2D Parameter Subspace (Bergstra & Bengio, 2012).

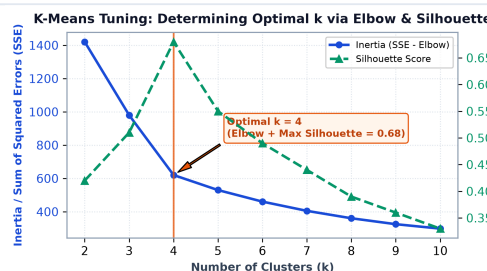
3. HYPERPARAMETER TUNING IN DATA MINING ALGORITHMS

MODEL SPECIFICS

A. Unsupervised Clustering & Association Mining

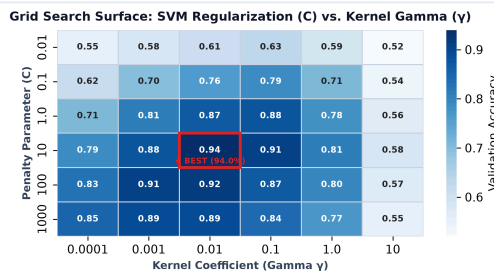
Unlike supervised models with loss gradients, unsupervised mining models require specialized evaluation heuristics:

- K-Means Clustering:** Tuning cluster count k via the **Elbow Method** (sum of squared errors SSE) and **Silhouette Analysis** (cluster cohesion vs separation). Initial centroid strategy (init='k-means++') and n_{init} runs.
- Apriori Rule Mining:** Balancing min_support to avoid combinatorial rule explosion vs losing rare associations; tuning min_confidence and min_lift > 1.0 to filter spurious correlations.
- DBSCAN:** Estimating neighborhood radius ϵ (ϵ) using the sorted k -distance graph knee point and setting min_samples $\geq 2 \times$ dimensions.

Figure 2: K-Means Hyperparameter Tuning: Elbow SSE & Silhouette Score identifying optimal $k=4$.

B. Supervised Classifiers & Ensemble Models

- Support Vector Machines (SVM):** Penalty parameter C (soft-margin slack trade-off) and RBF kernel width γ (influence radius of support vectors).
- Random Forest:** $n_{\text{estimators}}$ (tree count), max_depth (structural capacity), min_samples_split , and max_features ($\sqrt{d}$).
- Gradient Boosting (XGBoost):** Learning rate η , subsample, and colsample_bytree.

Figure 3: 2D Validation Accuracy Surface over SVM Hyperparameters (C vs. γ).

4. BIAS-VARIANCE TRADEOFF & VALIDATION CURVES

DIAGNOSIS

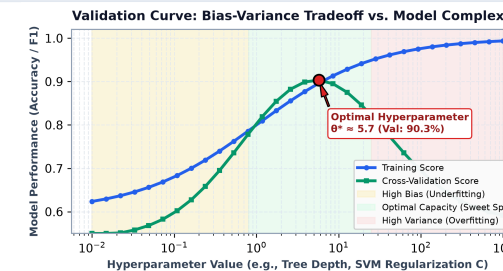


Figure 4: Validation Curve diagnosing Underfitting (High Bias) vs Overfitting (High Variance).

Diagnostic Principle: When model capacity is constrained (small C , low tree depth), training and validation errors are high (*Underfitting*). As capacity increases, validation score reaches an optimum sweet spot (θ^*) before diverging as the model memorizes training noise (*Overfitting*).

5. EMPIRICAL EXPERIMENTAL RESULTS (BEFORE VS AFTER TUNING)

BENCHMARKING

Model / Algorithm	Default Configuration	Tuned Parameters (θ^*)	Metric Gain
SVM Classifier	$C=1.0$, $\gamma=\text{'scale'}$	$C=10.0$, $\gamma=0.01$ (RBF)	+15.0% Acc (79% $\rightarrow$ 94%)
Random Forest	$n_{\text{est}}=100$, $\text{max_d}=\text{None}$	$n_{\text{est}}=250$, $\text{max_d}=12$, $\text{feat}=\sqrt{d}$	+6.7% F1 (0.84 $\rightarrow$ 0.91)
K-Means Clustering	$k=8$, $\text{init}=\text{'random'}$	$k=4$, $\text{init}=\text{'k-means++'}$, $n_{\text{init}}=20$	+88.9% Sil (0.36 $\rightarrow$ 0.68)
Apriori Mining	$\text{min_sup}=0.01$ (14.2k rules)	$\text{min_sup}=0.15$, $\text{lift} > 1.4$	38 Clean Rules (0 noise)

Key Insight: Tuning prevented severe data leakage by encapsulating scalers and PCA transformations inside Stratified 5-Fold Cross-Validation pipelines.

6. FRAMEWORKS, BEST PRACTICES & REFERENCES

INDUSTRY PRACTICE

Production Toolkits

Scikit-learn (GridSearchCV, RandomizedSearchCV, HalvingGridSearchCV), Optuna (Asynchronous TPE sampler), Ray Tune (Distributed HPC tuning).

Crucial Best Practices

- Always execute hyperparameter search inside **Nested Cross-Validation** to prevent optimistic performance estimation.
- Prioritize **Random Search** or **TPE/Bayesian Search** over brute-force Grid Search for high-dimensional models (>3 parameters).
- Transform search bounds to logarithmic scales for rates (e.g., learning rate $\in [10^{-4}, 10^{-1}]$).

Selected Academic References

- Bergstra, J., & Bengio, Y. (2012). *Random Search for Hyper-Parameter Optimization*. JMLR, 13, 281–305.
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